

Análisis Student Exam Results

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1 Introducción

El presente análisis explora el dataset *Student Exam Results*, el cual contiene información de **1000 estudiantes** y sus calificaciones en seis materias:

- Hindi
- English
- Science
- Maths
- History
- Geograpgy

Además, incluye tres columnas adicionales:

- **Total:** suma de las seis materias (se elimina por ser redundante)
- **Results:** 0 = reprobado, 1 = aprobado
- **Div:** categorías académicas 0 a 3

El propósito es realizar:

- Exploración descriptiva
 - Visualización
 - Análisis de correlación
 - Análisis de Componentes Principales (PCA)
-

2 Carga de Librerías y Datos

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.6
## ✓ forcats    1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.1      ✓ tibble     3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.2.0
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggcorrplot)
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
df <- read.csv("Student_Exam_Results.csv")
```

3 Limpieza del Dataset

Se elimina la columna Total y se convierten a factor las variables categóricas.

```
df1 <- df %>%
  select(-Total) %>%
  mutate(
    Results = as.factor(Results),
    Div = as.factor(Div)
  )
summary(df1)
```

##	Hindi	English	Science	Maths
##	Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
##	1st Qu.:26.00	1st Qu.:26.00	1st Qu.:25.00	1st Qu.:25.75
##	Median :53.00	Median :50.50	Median :50.00	Median :49.00
##	Mean :51.65	Mean :50.11	Mean :49.44	Mean :49.55
##	3rd Qu.:77.00	3rd Qu.:75.00	3rd Qu.:73.25	3rd Qu.:74.00
##	Max. :99.00	Max. :99.00	Max. :99.00	Max. :99.00
##	History	Geograpgy	Results Div	
##	Min. : 0.00	Min. : 0.00	0:650	0: 95
##	1st Qu.:24.00	1st Qu.:26.00	1:350	1:206
##	Median :49.00	Median :49.00		2:447
##	Mean :49.03	Mean :50.03		3:252
##	3rd Qu.:73.25	3rd Qu.:75.00		
##	Max. :99.00	Max. :99.00		

4 Distribución de Calificaciones por Materia

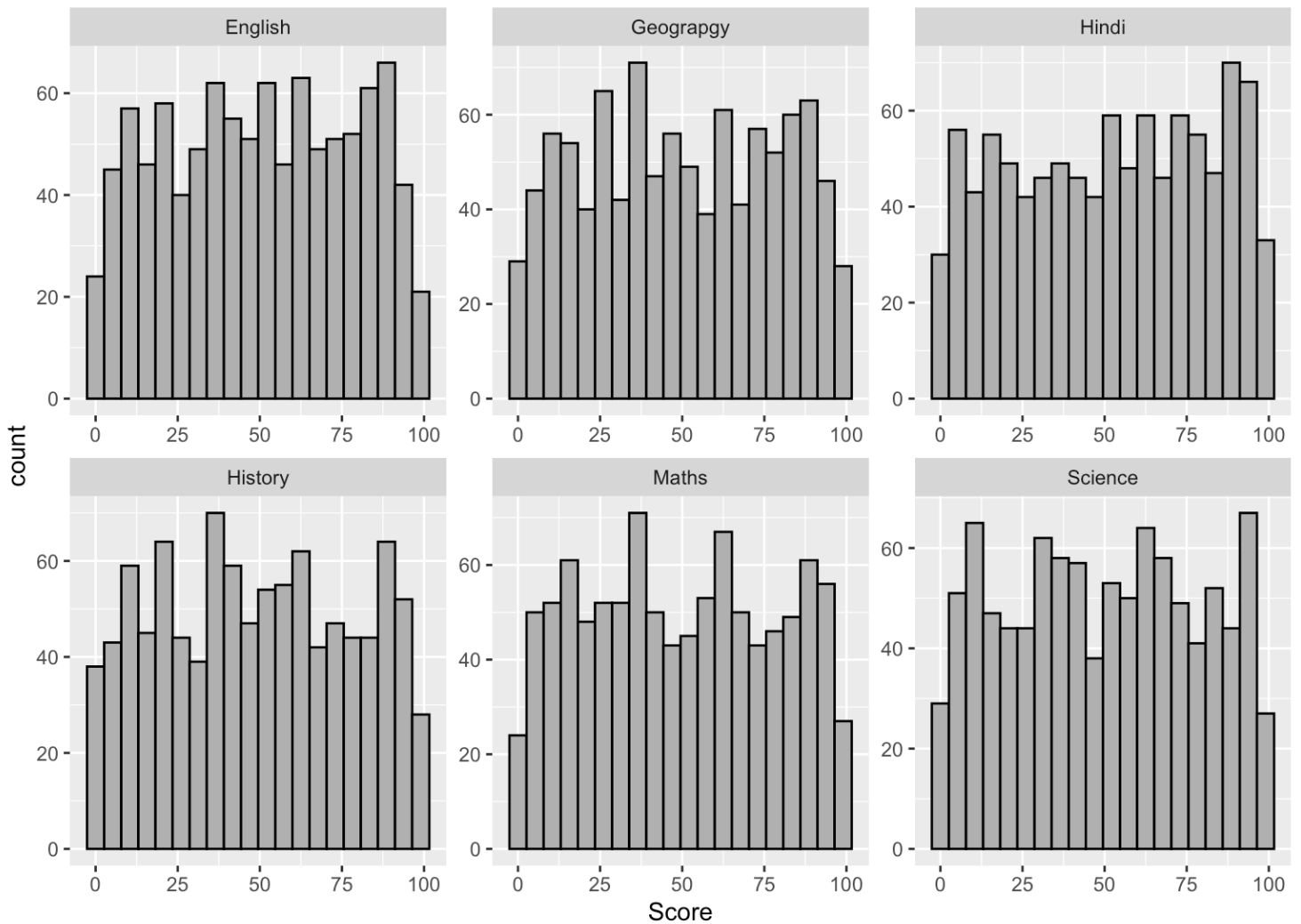
A continuación, se visualizan los histogramas de cada materia para observar su dispersión.

```
materias <- c("Hindi", "English", "Science", "Maths", "History", "Geograpgy")

df1 %>%
  pivot_longer(cols = materias, names_to = "Materia", values_to = "Score") %>%
  ggplot(aes(x = Score)) +
  geom_histogram(bins = 20, fill="grey70", color="black") +
  facet_wrap(~Materia, scales="free") +
  labs(title = "Distribución de Calificaciones por Materia")
```

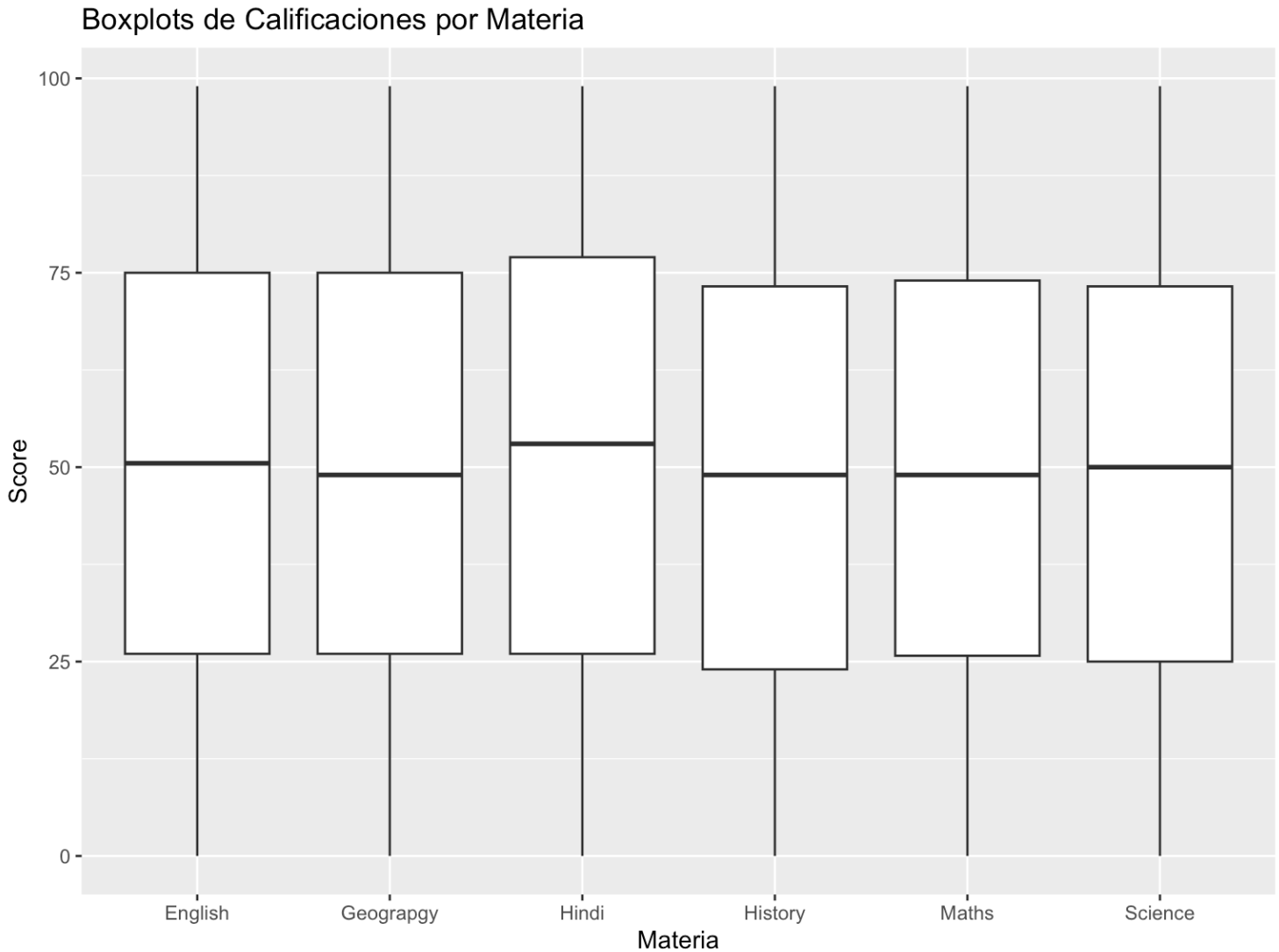
```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
## # Was:
## data %>% select(materias)
##
## # Now:
## data %>% select(all_of(materias))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Distribución de Calificaciones por Materia



5 Boxplots por Materia

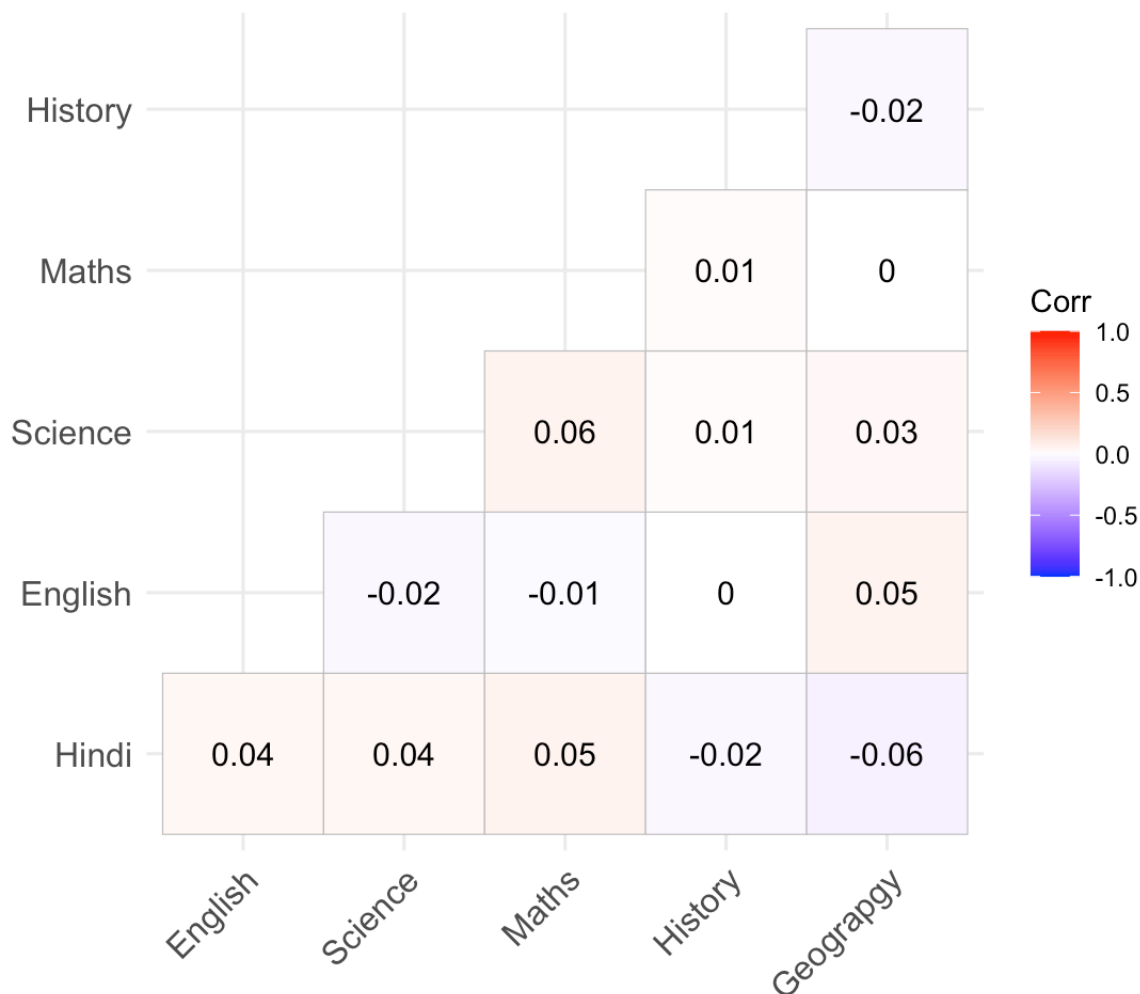
```
df1 %>%
  pivot_longer(cols = materias, names_to = "Materia", values_to = "Score") %>%
  ggplot(aes(x = Materia, y = Score)) +
  geom_boxplot() +
  labs(title = "Boxplots de Calificaciones por Materia")
```



6 Matriz de Correlación

```
corr_mat <- cor(df1[materias], method = "spearman")
ggcorrplot(corr_mat, type = "lower", lab = TRUE)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the ggcorrplot package.
## Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Las correlaciones entre materias son bajas, lo que indica que el desempeño en una materia no predice fuertemente el desempeño en otra.

7 Análisis PCA

Las materias se estandarizan antes de aplicar PCA.

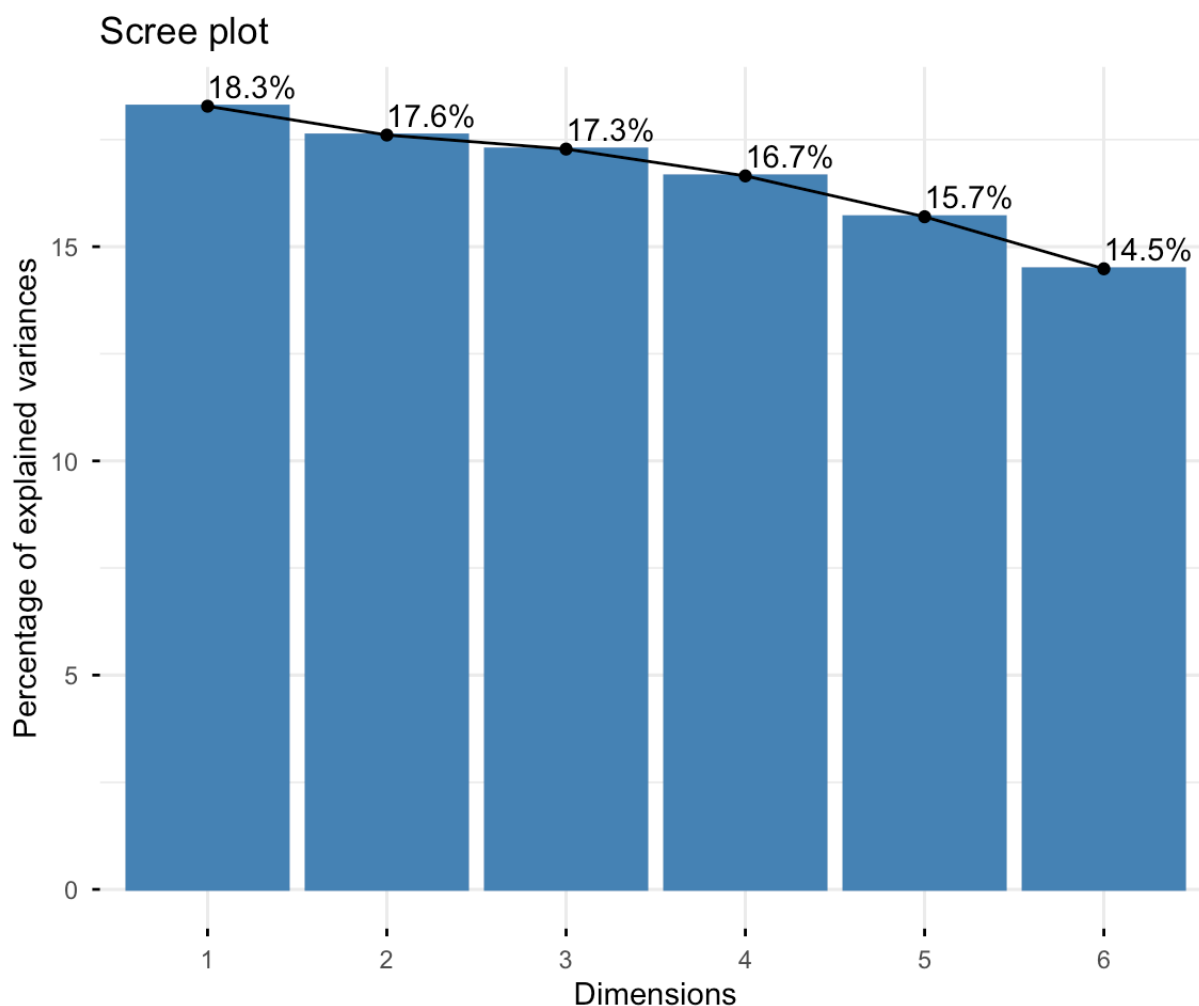
```
materias_scaled <- scale(df1[materias])
pca_res <- prcomp(materias_scaled, center = TRUE, scale. = TRUE)
summary(pca_res)
```

```
## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation  1.0473 1.0278 1.0182 0.9996 0.9705 0.9322
## Proportion of Variance 0.1828 0.1761 0.1728 0.1665 0.1570 0.1448
## Cumulative Proportion 0.1828 0.3589 0.5316 0.6982 0.8552 1.0000
```

8 Scree Plot

```
fviz_eig(pca_res, addlabels = TRUE)
```

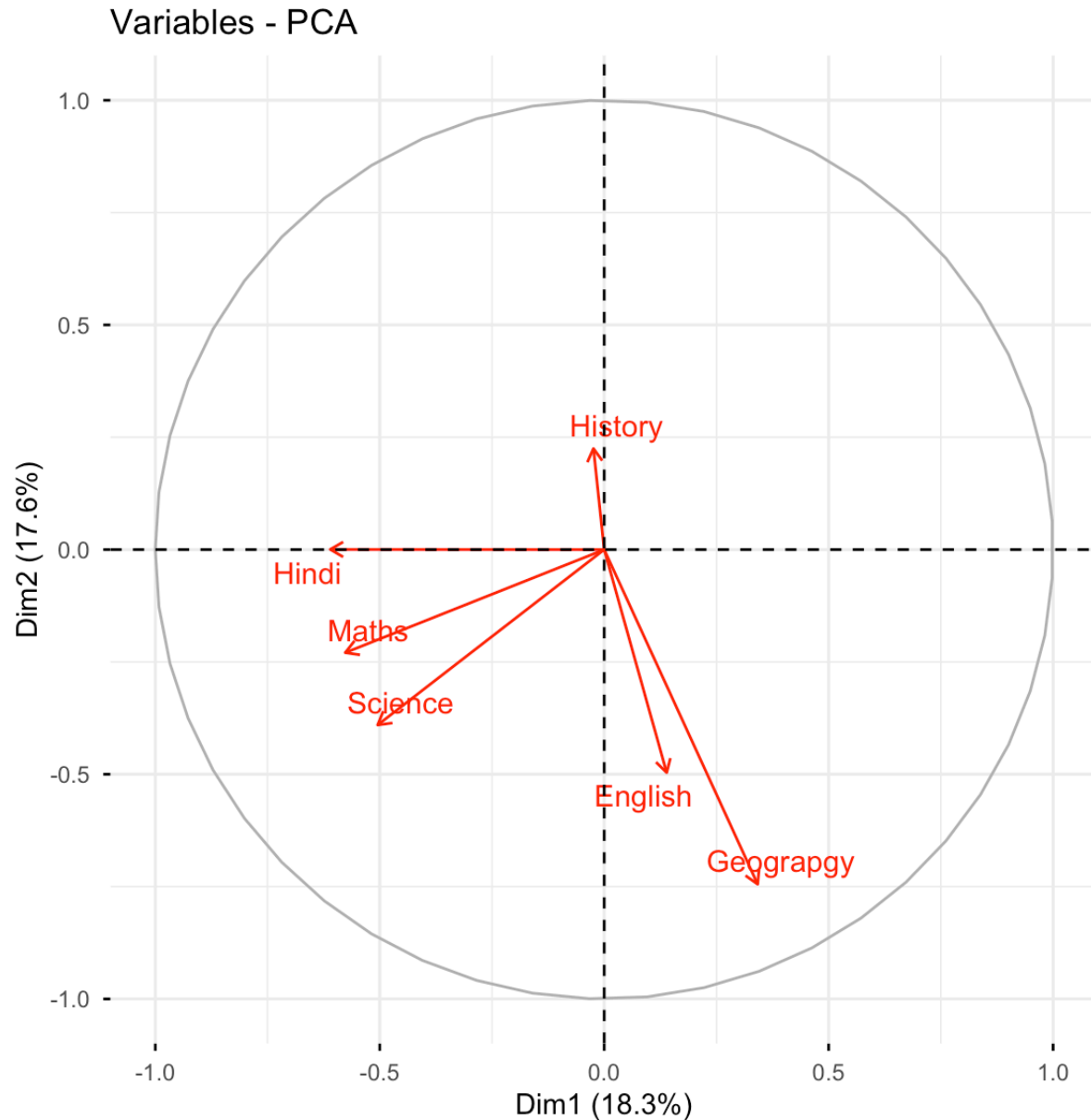
```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :  
## Ignoring empty aesthetic: `width`.
```



9 Biplot de Variables

```
fviz_pca_var(pca_res, repel = TRUE, col.var = "red")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## i The deprecated feature was likely used in the ggpubr package.  
## Please report the issue at <https://github.com/kassambara/ggpubr/issues>.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



10 Conclusiones

- Las materias presentan correlación baja, lo que limita la capacidad de PCA para reducir la dimensión

del dataset.

- Las dos primeras componentes principales explican alrededor del 35% de la varianza total.
- El PCA funciona como herramienta exploratoria, pero no logra resumir eficientemente la información de las seis materias.